

Rishi Raj

Doctoral Researcher in Mathematical Physics • Aspiring Software Developer

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TECHNICAL SKILLS

Programming Languages: Proficient in Python (scientific and production-grade), C/C++ (high-performance computing) with CMake, JavaScript / TypeScript (full-stack systems)

ML & AI Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face, Transformers, Weights & Biases (wandb), Pandas visualization with Seaborn, R, ROOT

Cloud & DevOps: End-to-end model deployment pipelines with AWS & GCP, Docker, CI/CD via GitHub Actions, ML model deployment & monitoring, MLOps best practices

Scientific Computing: Mathematica, GNU Scientific Library (GSL), Boost C++, Julia, SageMath

Dev Tools & Version Control: Advanced Git workflows, Linux System Administration, VS Code

Full Stack Web Development: Node.js, Flask, FastAPI, React based frameworks like Next.js, MongoDB, SQL

Core Competencies: Research-grade problem solving, Continuous Learning, Collaborative Problem-Solving, Cross-Functional Communication, Leadership, Adaptability

Langauges: English (native), French (beginner), German (beginner, actively pursuing a langauge course)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

May 2023 – Jul 2023

Supervised by Boris Pioline @ LPTHE, Sorbonne Université — Paris, France

- Utilized Mathematica to study bound states of BPS particles in the Coulomb branch of $\mathcal{N} = 4$ supersymmetric quiver quantum mechanics (QQM), revealing structural features of the moduli space.
- Performed a comparative analysis of BPS index computation via attractor flow trees, Coulomb branch localization, and modular anomaly methods, offering insights into wall-crossing behavior.
- Helped clarify the contribution of scaling solutions using the Minimum Modification Hypothesis (MMH), reproducing attractor flow tree results from the Coulomb branch formula.
- Applied graph-theoretic and numerical methods to compute physical invariants, improving precision and performance in multi-centered black hole configurations.

Graduate Research Assistant

May 2023 – Jul 2023

Supervised by Ayan Mukhopadhyay @ IIT Madras — Chennai, India

- Utilized Mathematica to study bound states of BPS particles in the Coulomb branch of $\mathcal{N} = 4$ supersymmetric quiver quantum mechanics (QQM), revealing structural features of the moduli space.
- Performed a comparative analysis of BPS index computation via attractor flow trees, Coulomb branch localization, and modular anomaly methods, offering insights into wall-crossing behavior.
- Helped clarify the contribution of scaling solutions using the Minimum Modification Hypothesis (MMH), reproducing attractor flow tree results from the Coulomb branch formula.
- Applied graph-theoretic and numerical methods to compute physical invariants, improving precision and performance in multi-centered black hole configurations.

Web Development Freelancer

Nov 2021 – Jul 2022

Zenoholics Pvt. Ltd. — Bangalore, India

- Lead the modernization of full-stack applications using React.js, Next.js, and MongoDB, achieving a 30% performance gain through architectural refinement and performance profiling.
- Leveraged advanced diagnostics tools—DevTools, Profiler API, Lighthouse, Postman, and Bundle Analyzer—to identify bottlenecks and guide optimization across the stack.
- Engineered robust CI/CD workflows with GitHub Actions, Docker, and AWS; incorporated Kubernetes for container orchestration and scalable deployments.
- Gained hands-on experience in scaling distributed systems via RabbitMQ, connection pooling, caching strategies, lazy loading, CDNs, and load testing, enhancing backend resilience and frontend responsiveness.

EDUCATION

Doctor of Philosophy, PhD in Mathematical Physics

Oct 2023 – Sept 2026 (expected)

Sorbonne Université, Paris, France

Research focus on *BPS Black Holes and Donaldson-Thomas Invariants* under the supervision of Boris Pioline.

Advanced Deep Learning (ongoing), Applied Machine Learning, String Theory, Quantum Field Theory II, Supersymmetry and Supergravity, Differential Geometry

Master of Science, MS in Theoretical Physics

Aug 2022 – Jul 2023

Indian Institute of Technology, IIT Madras, Chennai, India

9.18/10 CGPA

Master's thesis on *Black Holes through the lens of Matrix Theory*.

Bachelor of Science, BS in Physics

July 2018 – Aug 2022

Indian Institute of Technology, IIT Madras, Chennai, India

9.18/10 CGPA

Graduated with a Minor in Mathematics

ACADEMIC PROJECTS

Free statistics in BFSS matrix model

Jan 2025 – Jul 2025

Ongoing work with Boris Pioline

- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Scattering of non-extremal supersymmetric Black Holes

Jan 2025 – Jul 2025

Ongoing work with Boris Pioline

- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Improving smuon searches with Neural Networks

Feb 2025 – Present

Ongoing work with Mark Goodsell et al.

- Investigating advanced machine learning architectures (**TabNet**, **Particle Flow Networks**, **XGBoost**) to enhance signal-background discrimination in smuon searches at the LHC.
- Working with Monte Carlo simulated datasets to develop robust preprocessing pipelines and feature representations for high-dimensional collider data.
- Exploring adjacent applications such as **detector unfolding** and **systematics-aware training**, aiming to improve interpretability and generalization to experimental data.
- Benchmarking model performance against traditional cut-based analyses to quantify gains in statistical sensitivity and background rejection.

Multi-centered Black Hole Quantum Mechanics and Generalized Error Functions

Jan 2025 – Jul 2025

Ongoing work with Boris Pioline

- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

BPS Dendroscopy on Local $\mathbf{P}^1 \times \mathbf{P}^1$

Mar 2024 – Dec 2024

- Developed neural networks to reconstruct bulk geometry from boundary physics in holographic setups.
- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Learning Holographic Horizons

Jan 2023 – Dec 2023

- Developed neural networks to reconstruct bulk geometry from boundary physics in holographic setups.
- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Black Holes through the lens of Matrix theory

Oct 2022 – April 2023

- Designed a simulation suite for classical and quantum simulations of the M-theory matrix model.
- Achieved performance improvements of up to 17x for simulations of black hole dynamics described by random matrix ensembles.
- Studied deviations in thermalization behavior to identify underlying quantum mechanical properties.

CONFERENCES AND SUMMER SCHOOLS

4EU+ Ph.D. Summer School on Machine Learning in Physics	June 2025
@ Niels Bohr Institute, University of Copenhagen — Copenhagen, Denmark	
Completed an intensive summer school — covered advanced ML topics including Boosted Decision Trees, Bayesian Neural Networks, Transformers, Anomaly Detection, and Domain Adaptation; trained and optimized a Shakespearean language model; evaluated by an independent committee through lecture participation, hands-on exercises, and model deployment on large structured datasets.	
Strings2024	Aug 2022 – Jul 2023
@ CERN — Geneva, Switzerland	9.18/10 CGPA
Master's thesis on <i>Black Holes through the lens of Matrix Theory</i> .	
1st ROOT Hackathon	July 2018 – Aug 2022
@ CERN — Geneva, Switzerland	9.18/10 CGPA
Graduated with a Minor in Mathematics	
CERN Winter School on Supergravity, Strings and Gauge Theory	July 2018 – Aug 2022
@ CERN — Geneva, Switzerland	9.18/10 CGPA
Graduated with a Minor in Mathematics	
LACES 2023	July 2018 – Aug 2022
@ Galileo Galilei Institute for Theoretical Physics — Florence, Italy	9.18/10 CGPA
Graduated with a Minor in Mathematics	
Solay Doctoral School on Quantum Field Theory, Strings and Gravity	July 2018 – Aug 2022
@ École Normale Supérieure (ENS) — Paris, France, and CERN — Geneva, Switzerland	9.18/10 CGPA
Graduated with a Minor in Mathematics	

AWARDS

64th Institute day – ‘Mr S. Venkitaramanan, I.A.S Retd’ prize	Aug 2023
IIT Madras — Chennai, India	
EDPIF Doctoral Fellowship	Jun 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
‘Electronics For You’ prize	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
Charpak 2023 Fellow	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
Alumni Travel Grant	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
NIUS 2019 Fellow	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
Microsoft Codefundo++	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	
KVPY Fellow	Jul 2023
Sorbonne University, École Doctorale Physique en Île-de-France (EDPIF) — Paris, France	

OTHER WORK EXPERIENCE

Teaching Assistant	Jul 2022 – Nov 2022
PH5060 (Physics Lab I), IIT Madras — Chennai, India	
• Assisted students with computational physics problems, including chaotic systems and Monte Carlo simulations. • Conducted viva sessions and evaluated reports to assess students' understanding and progress.	

Horizon Club, IIT Madras

Sep 2019

Delivered a lecture series titled *Relativity from Symmetries*, introducing group theory concepts and special relativity to physics enthusiasts

- Designed and presented four lectures focusing on symmetry principles, Lorentz transformations, and physical implications of special relativity.
- Engaged and inspired beginning physics enthusiasts to explore advanced theoretical concepts through interactive discussions.

Music Club, IIT Madras

Jul 2019 – Nov 2019

Organized and coordinated musical events, including performances for the annual cultural festival, Saarang

- Led the planning and execution of multiple musical events, ensuring seamless coordination among performers and technical teams.
- Enhanced event management and teamwork skills while handling responsibilities for the institute's cultural programs.